

**SCHOOL OF
COMPUTING**

Amrita Vishwa Vidyapeetham, Chennai Campus
Department of Computer Science and Engineering
Exercise-6: PCA

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CLASS : CSE-B

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Question: For the below dataset, identify which variable is principal component. | Variable | 1 | 2 | 3 | | ----- | ----: | -----: | -----: | | Climate | 0.190 | 0.017 | 0.207 | | Housing | 0.544 | 0.020 | 0.204 | | Health | 0.782 | -0.605 | 0.144 | | Crime | 0.365 | 0.294 | 0.585 | | Transportation | 0.585 | 0.085 | 0.234 | | Education | 0.394 | -0.273 | 0.027 | | Arts | 0.985 | 0.126 | -0.111 | | Recreation | 0.520 | 0.402 | 0.519 | | Economy | 0.142 | 0.150 | 0.239 |

```
import pandas as pd
import numpy as np

data = {
    'Variable': [
        'Climate',
        'Housing',
        'Health',
        'Crime',
        'Transportation',
        'Education',
        'Arts',
        'Recreation',
        'Economy'
    ],
    'PC1': [
        0.190,
        0.544,
        0.782,
        0.365,
        0.585,
        0.394,
        0.985,
        0.520,
        0.142
    ],
    'PC2': [
        0.017,
        0.020,
        -0.605,
        0.294,
        0.085,
        -0.273,
        0.126,
        0.402,
        0.150
    ],
    'PC3': [
        0.207,
        0.204,
```

```

        0.144,
        0.585,
        0.234,
        0.027,
        -0.111,
        0.519,
        0.239
    ]
}

df = pd.DataFrame(data)

print(df)

```

	Variable	PC1	PC2	PC3
0	Climate	0.190	0.017	0.207
1	Housing	0.544	0.020	0.204
2	Health	0.782	-0.605	0.144
3	Crime	0.365	0.294	0.585
4	Transportation	0.585	0.085	0.234
5	Education	0.394	-0.273	0.027
6	Arts	0.985	0.126	-0.111
7	Recreation	0.520	0.402	0.519
8	Economy	0.142	0.150	0.239

```

pc1_index = df['PC1'].abs().idxmax()

print("PC1 variable:", df.loc[pc1_index, 'Variable'])
print("PC1 loading:", df.loc[pc1_index, 'PC1'])

PC1 variable: Arts
PC1 loading: 0.985

pc2_index = df['PC2'].abs().idxmax()

print("PC2 variable:", df.loc[pc2_index, 'Variable'])
print("PC2 loading:", df.loc[pc2_index, 'PC2'])

PC2 variable: Health
PC2 loading: -0.605

pc3_index = df['PC3'].abs().idxmax()

print("PC3 variable:", df.loc[pc3_index, 'Variable'])
print("PC3 loading:", df.loc[pc3_index, 'PC3'])

PC3 variable: Crime
PC3 loading: 0.585

for pc in ['PC1', 'PC2', 'PC3']:
    index = df[pc].abs().idxmax()

```

```

print(
    pc,
    "->",
    df.loc[index, 'Variable'],
    "-> Loading:",
    df.loc[index, pc]
)

PC1 -> Arts -> Loading: 0.985
PC2 -> Health -> Loading: -0.605
PC3 -> Crime -> Loading: 0.585

result = []

for pc in ['PC1', 'PC2', 'PC3']:
    index = df[pc].abs().idxmax()

    result.append([
        pc,
        df.loc[index, 'Variable'],
        df.loc[index, pc]
    ])

result_df = pd.DataFrame(
    result,
    columns=['Principal Component', 'Variable', 'Loading']
)

print(result_df)

```

	Principal Component	Variable	Loading
0	PC1	Arts	0.985
1	PC2	Health	-0.605
2	PC3	Crime	0.585

```

import matplotlib.pyplot as plt

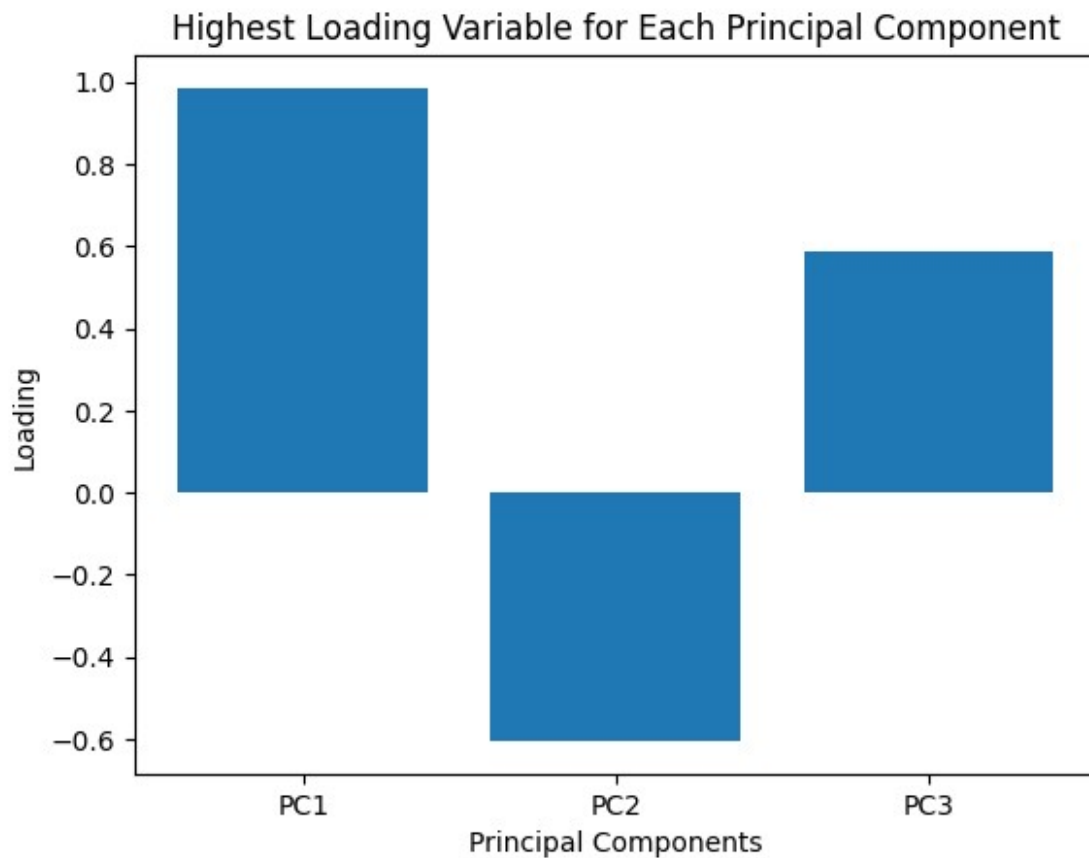
variables = ['Arts', 'Health', 'Crime']
loadings = [0.985, -0.605, 0.585]

plt.bar(['PC1', 'PC2', 'PC3'], loadings)

plt.xlabel('Principal Components')
plt.ylabel('Loading')
plt.title('Highest Loading Variable for Each Principal Component')

plt.show()

```



Thus, the principal components were identified based on the highest absolute loading values. PC1 corresponds to Arts (0.985), PC2 corresponds to Health (-0.605), and PC3 corresponds to Crime (0.585).